Checkout Policy — ADR-006 Supplement

Version 1.0 | Status: Approved policy / Documentation review

Retention

Completed checkout idempotency records must be retained for at least 24 hours after completion. After expiration, deduplication of old requests is not guaranteed.

Replay

A retry by the same authenticated customer with the same key and equivalent canonical request payload returns the stored result and must not create a second order.

Payload mismatch

Reusing a key with a different canonical request payload returns HTTP 409 Conflict without changing order or inventory state.

Concurrent request

Only one in-flight request per customer and idempotency key may execute checkout. Concurrent duplicates receive HTTP 409 Conflict with a retryable error response; the client should retry after the first request completes.

Cancellation

Only CONFIRMED orders may transition to CANCELLED. The cancellation and stock restoration execute atomically and stock must be restored exactly once.

Failure recovery

Idempotency state changes and order creation must be coordinated transactionally. Incomplete operations must have a defined recovery path and cannot be replayed as completed orders.

Mock payment

The mock payment service does not process real funds or collect card details. A failed mock payment creates no order and changes no inventory.

Acceptance Tests

- TC-IDEM-001: Replaying an identical successful checkout returns the original order and does not decrement stock again.
- TC-IDEM-002: Reusing a key with a different payload returns 409 Conflict.
- TC-IDEM-003: Two simultaneous requests with the same key do not create duplicate orders.
- TC-IDEM-004: Retention of a completed idempotency record is at least 24 hours.
- TC-CANCEL-001: Cancelling a CONFIRMED order restores its stock exactly once.
- TC-CANCEL-002: Repeated cancellation is rejected and does not increase stock.

- TC-CONC-001: Two customers purchasing the final stock unit result in exactly one successful order.
- TC-PAY-001: Mock payment failure leaves cart, order and inventory unchanged.